

SpecTantra AI: Low-Cost Smartphone Soil Testing & Fertilizer Check for Indian Farmers

Project Title:

SpecTantra AI

Ultra-Low-Cost Smartphone Spectrometry for Soil Diagnostics & Fertilizer Authentication

- **Focused Areas:** Smart Agriculture & Rural Innovation / Sustainability & Environment
- **Target Region:** Pan-India (All agricultural states across India)
- **Submission Category:** Software-Hardware Synergy (Frugal Engineering / "Jugaad" Innovation)

1. Executive Summary

SpecTantra AI is a simple, offline mobile and computer software tool. It helps small farmers all over India test their soil and check if their fertilizers are pure, using tools they already own.

Instead of buying expensive laboratory equipment, a farmer takes a tiny $1.5\text{ cm} \times 1.5\text{ cm}$ slice of an old, discarded CD or DVD and tapes it over their smartphone's camera. This clear plastic piece acts like a **glass prism**. When light bounces off a soil or fertilizer sample and passes through this CD piece, it splits into a beautiful **rainbow strip** on the phone's camera.

Our smart software reads this rainbow. By looking at which colors are bright and which colors are dark (absorbed), the software can instantly estimate the levels of **Nitrogen** (N), **Phosphorus** (P), and **Potassium** (K) in the soil. It can also detect if the fertilizer has been mixed with cheap, fake materials like sand or gypsum.

Because the software works 100% offline and speaks to the farmer in their local regional languages (via voice messages), it brings advanced science directly to Indian fields at almost **zero hardware cost**.

2. Project Objectives

Our project has four clear goals to help Indian agriculture:

- **Save Money and Time:** Traditional laboratory soil tests take **15 days** and cost between **Rs. 1000 to Rs. 2000** . Our tool gives instant results on the field and costs less than **Rs. 10** to set up.
- **Stop Problems Before They Happen:** Most mobile apps only find plant diseases after the leaves turn yellow and the crop is already damaged. Our tool tests the soil *before* planting, preventing crop loss entirely.
- **Clean up E-Waste:** We reuse old, discarded CDs and DVDs that would otherwise end up in landfills, turning trash into a scientific tool.
- **Work Anywhere (Offline):** Remote farms in India often have no internet or mobile network. Our app runs completely offline, with local language voice guides to help every farmer.

3. The Problem

The Soil Testing Gap

India has millions of small farms, but very few soil testing laboratories. Farmers have to travel far, pay high fees, and wait weeks for test results. Because of these delays, they often plant crops without knowing what nutrients their soil actually needs.

Fake and Adulterated Fertilizers

Due to shortages and illegal supply chains, high-demand fertilizers like Urea or DAP are often mixed with cheap fillers like gypsum, salt, clay, or industrial sand. This fake fertilizer does not feed the crops, wastes the farmer's hard-earned money, and permanently damages the soil's chemistry.

The Limits of Current Apps

Most farming mobile apps today just take regular photos of plant leaves. But by the time a plant's leaves show damage from lack of nutrients, the crop's yield has already dropped by 30% to 50%. Also, standard phone cameras can only see three broad colors: Red, Green, and Blue. They cannot see the highly specific details needed to detect actual chemical compounds.

4. How the Science Works

The Magic of the CD/DVD (Light Splitter)

A CD or DVD looks smooth, but it actually has thousands of microscopic, parallel tracks pressed into its plastic base.

- A standard CD has tracks spaced about $1.6\ \mu\text{m}$ apart.
- A DVD has even tighter tracks, about $0.74\ \mu\text{m}$ apart.

These tiny tracks act as a **diffraction grating** (which works exactly like a glass prism). When light reflected from our soil sample passes through the slit and the clear CD piece, different colors of light bend at different angles based on their wavelength (λ).

The formula for this split is:

$$d \sin(\theta) = m\lambda$$

Where:

- d is the microscopic track spacing on the CD/DVD,
- θ is the angle at which the light bends,
- m is the order of the light band,
- λ is the color wavelength (e.g., $450\ \text{nm}$ for blue, $650\ \text{nm}$ for red).

This physical reaction projects a perfect, continuous rainbow strip directly onto the camera sensor, allowing us to see every single-color wavelength from $400\ \text{nm}$ (violet) to $700\ \text{nm}$ (red).

Chemical Wavelength Fingerprints

Every element in nature absorbs specific colors of light.

- **Nitrogen (N)**: Absorbs specific wavelengths in the blue ($430\ \text{nm} - 470\ \text{nm}$) and deep-red ($640\ \text{nm} - 680\ \text{nm}$) spectrum.
- **Phosphorus (P) & Potassium (K)**: Create distinct light-absorption patterns when mixed with simple, low-cost organic color-changing liquid kits.
- **Fertilizer Purity**: Pure chemical fertilizers reflect bright, sharp light peaks. If sand, gypsum, or salt is mixed in, these peaks disappear or flatten out, letting our software know the sample is fake.

5. Software Plan

Our project is designed to grow in two simple stages:

Phase 1: Python/OpenCV Desktop Prototype (Current Working Model)

To prove the science works quickly, we built a working computer program:

1. **Camera Feed:** The computer connects to a smartphone camera over local Wi-Fi or USB.
2. **On-Screen Guide:** The program draws a blue box on the screen. The user aligns the rainbow strip inside this box.
3. **Measuring Brightness:** The software crops the rainbow image, converts it to grayscale, and calculates the average brightness of each column of pixels from left to right:

$$\text{Intensity}(x) = \frac{1}{H} \sum_{y=0}^{H-1} \text{PixelValue}(x, y)$$

4. **Live Graphing:** It plots a live graph on the screen. Whenever a nutrient is present, a sharp dip or valley appears on the graph.

Phase 2: Offline Android Mobile App (Future Scale-up)

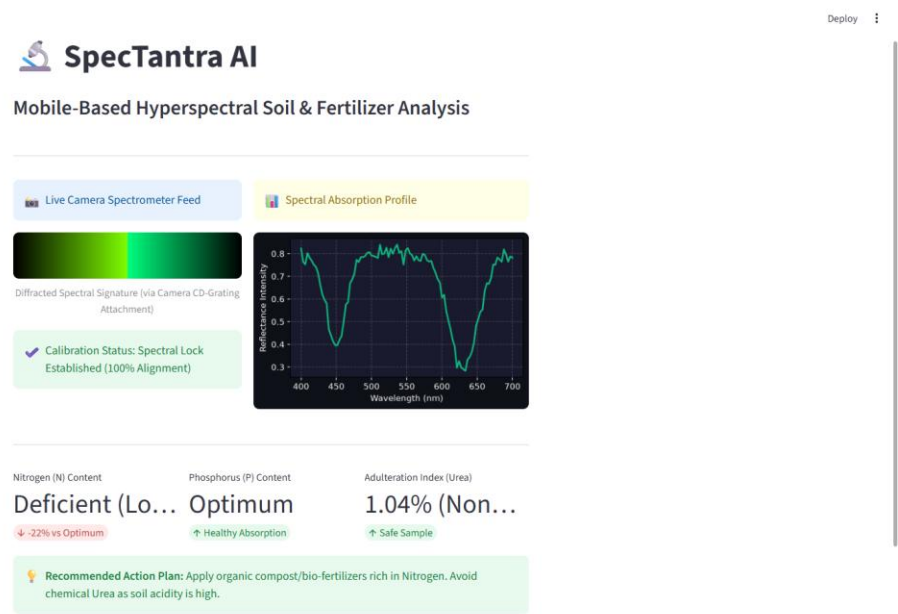
Next, we will package this code into an easy-to-use Android App:

- **Camera Lock:** The app will lock the phone's auto-exposure, focus, and white balance so changing room lights won't interfere with the measurements.
- **On-Device AI:** A lightweight machine learning model (using **TensorFlow Lite**) will run directly on the phone's processor to analyze the graph in less than **0.05 seconds** without needing any internet.

6. How a Farmer Uses It

1. **Prepare the Piece:** The farmer takes an unwanted CD/DVD, peels the shiny metallic layer off using tape, and cuts out a small, clear plastic square.
2. **Attach to Phone:** They tape this clear square over their smartphone's back camera.
3. **Set Up the Sample:** They place a small spoonful of dry soil or fertilizer under a bright white LED light.
4. **Scan:** They open the SpecTantra AI app. An on-screen box guides them: *"Align the rainbow inside the box."*
5. **AI Analysis:** The app instantly captures the image, analyzes the colors, and processes it offline.
6. **Voice Guidance:** Within seconds, the phone speaks the results aloud in their local regional language:
 - *Example (Hindi):* "आपके खेत की मिट्टी में नाइट्रोजन की मात्रा कम है। कृपया जैविक खाद का उपयोग करें।" (*Your soil is low on Nitrogen. Please apply organic compost.*)

7. Screenshot of the Prototype



8. Alignment with Lenovo's Vision: "Smarter Technology for All"

SpecTantra AI perfectly matches Lenovo's mission of "Smarter Technology for All":

- **Tech for Everyone:** Instead of requiring poor farmers to buy expensive machinery, we leverage the cheap Android phones they already own.
- **Saving the Environment:** This project upcycles plastic e-waste (old CDs) to save agricultural soils from chemical over-fertilization.
- **No Barriers:** By keeping everything ^{100%} offline and using regional voice outputs, we ensure that even semi-literate farmers in the most remote corners of India can benefit from artificial intelligence and advanced science.